solving problems with computers

digitalisierung und programmierung · prof. dr. nicolas meseth

1. what happens in the middle?
2. the boxes in this project
3. boxes all the way down
4. two questions inside every box

part 1

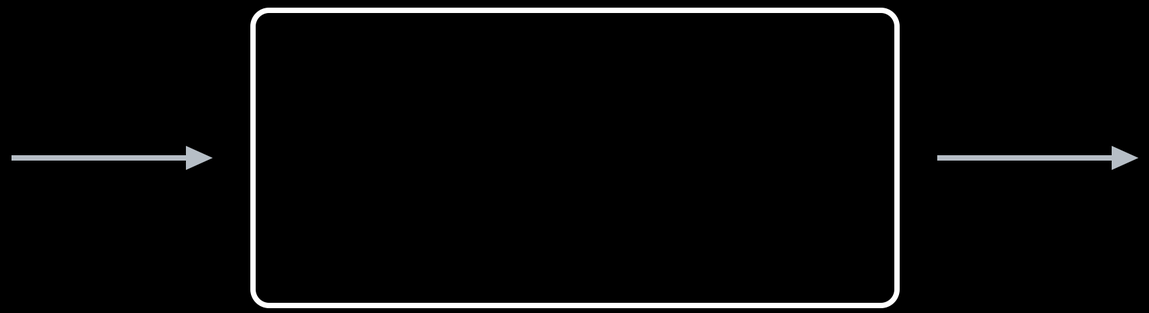
what happens in the middle?

you order.
a tray appears.
what happened
in between?

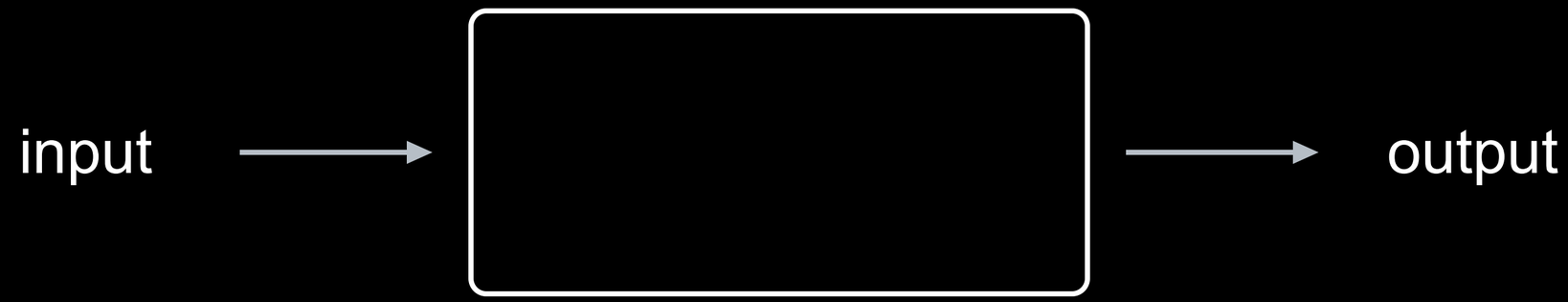


image: ai-generated (gpt-image-2)

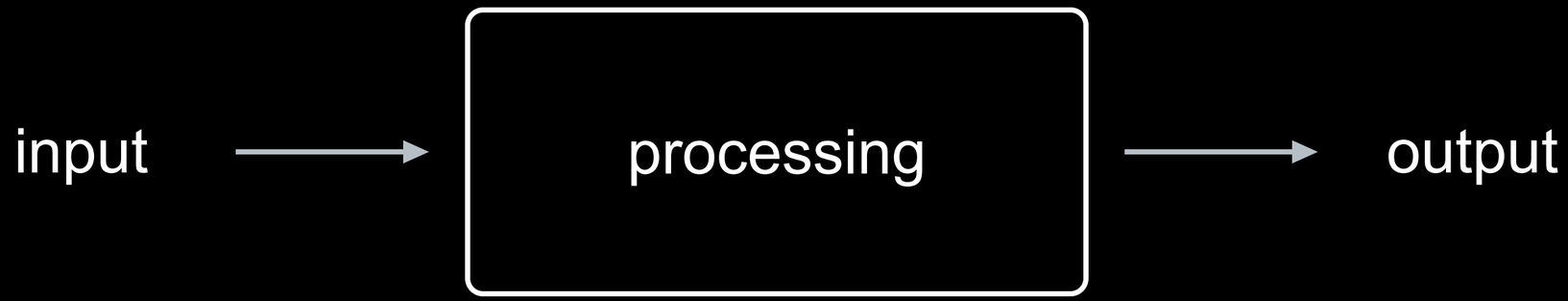
every computable piece has this shape



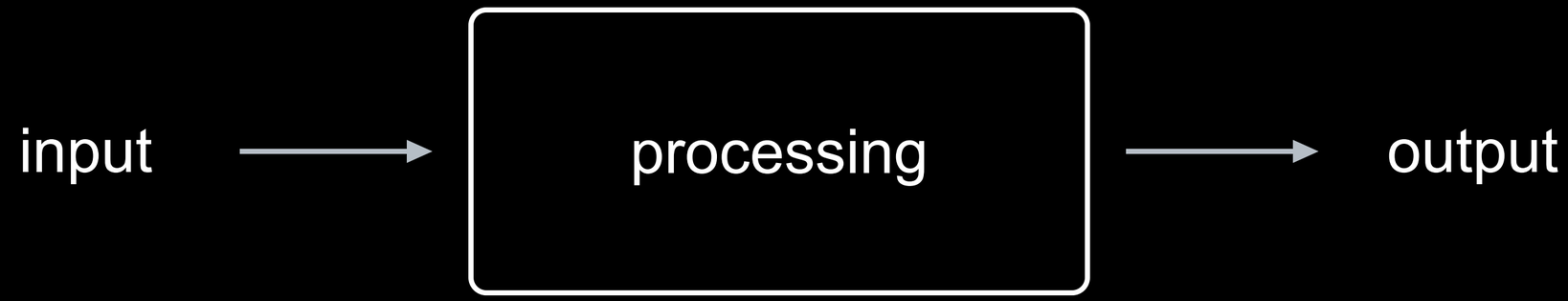
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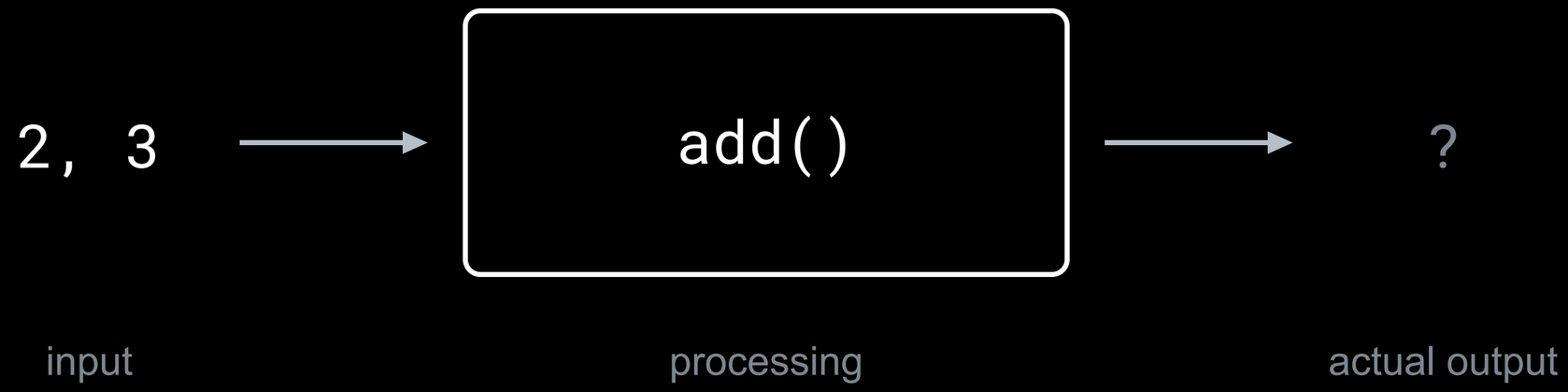
every computable piece has this shape



input · processing · output

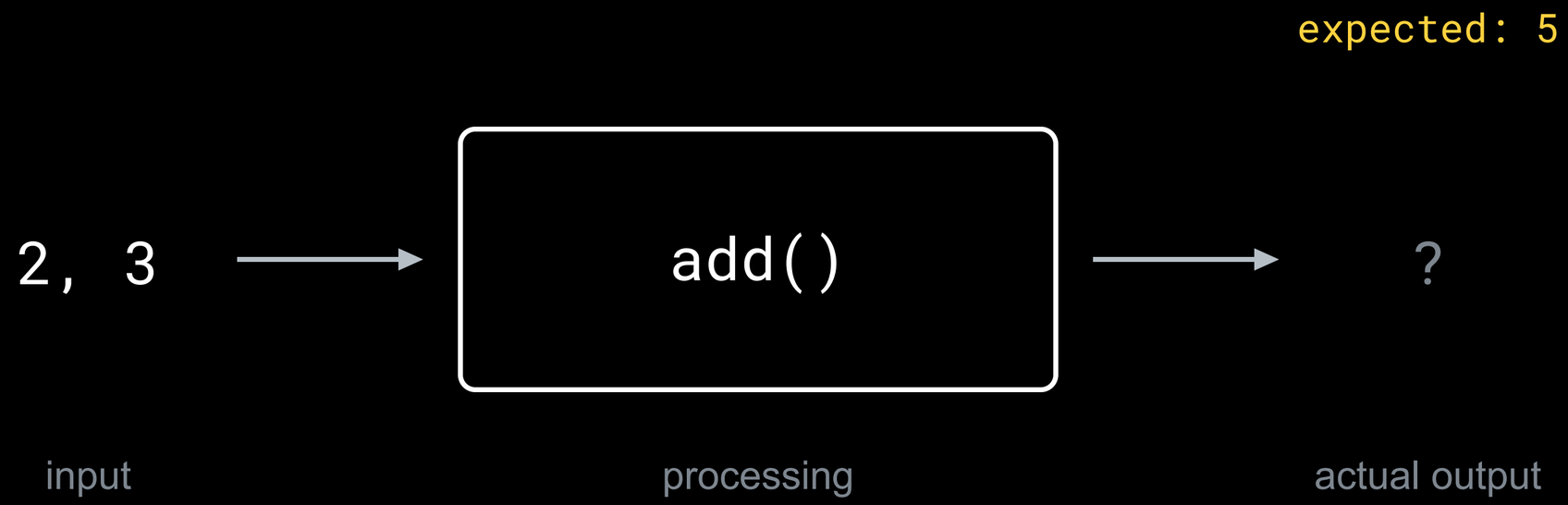
IPO

a box with clear edges brings its own test



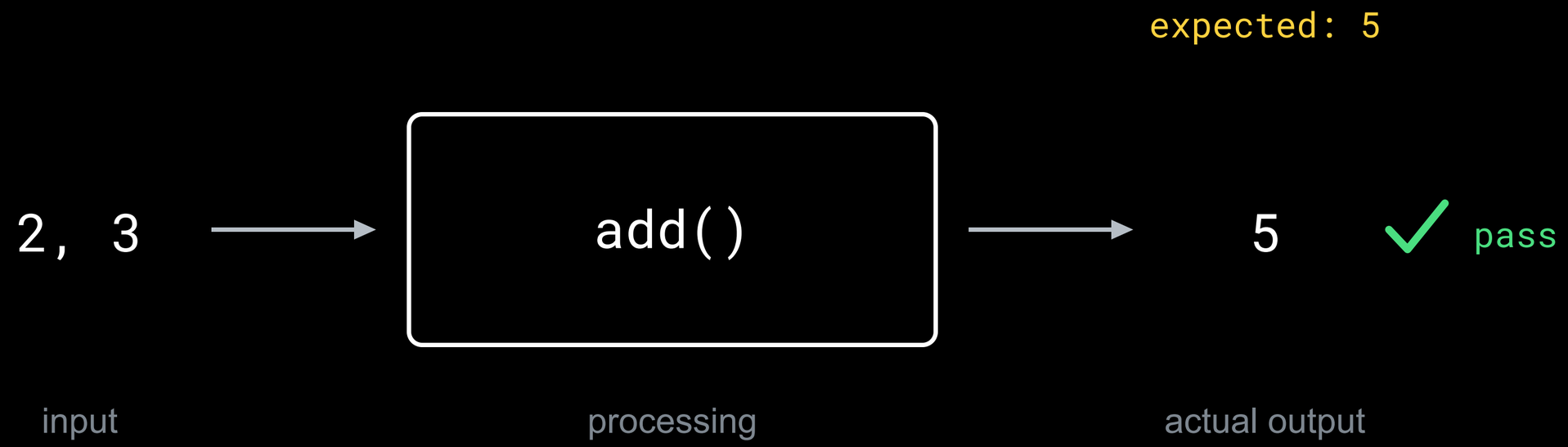
if you know the input and expected output, you have a test.

a box with clear edges brings its own test



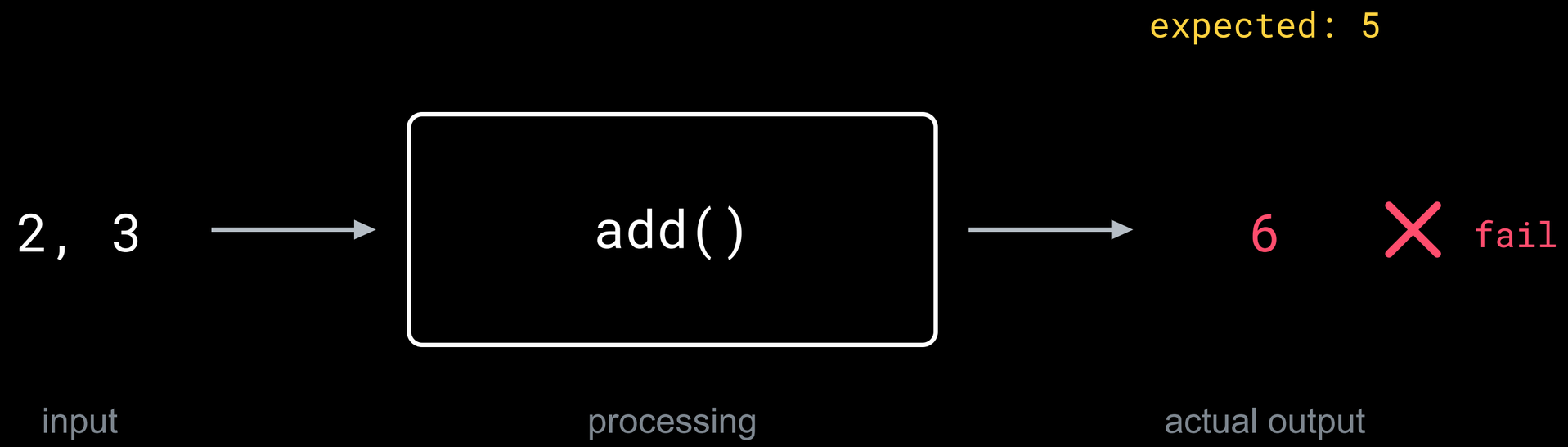
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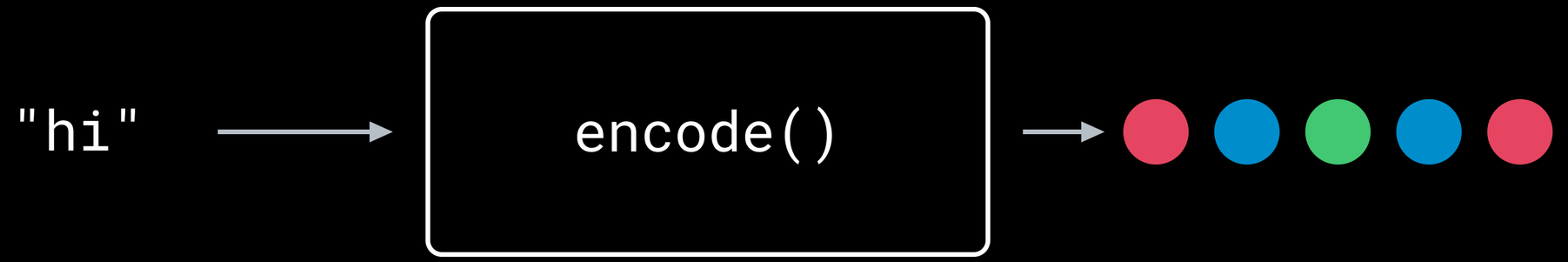
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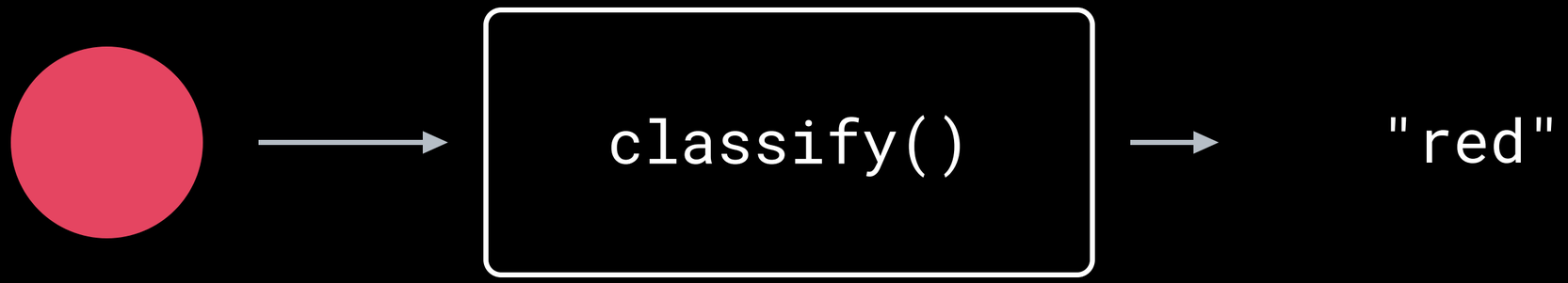
if you know the input and expected output, you have a test.

part 2

the boxes in this project

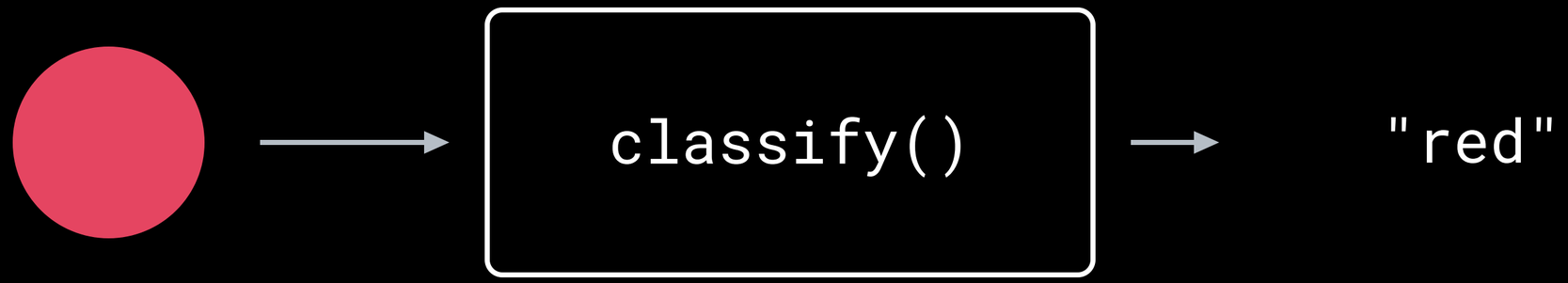

"hi"
encode()
text
program
colours

what enters the receiver?



your eyes see red. the box sees four numbers.

what enters the receiver?

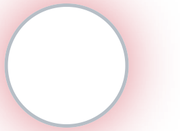


your eyes see red. the box sees four numbers.

try it: what does the sensor see?

sensor case: `clear red` | `weak red` | `mixed light` | `darkness`

1 input



R	●———	203
G	——●———	41
B	———●———	57
C	——————●—	310

→

2 processing

```
classify(r, g, b, c)

if c < min light → off
if lead < min lead → uncertain
else → strongest channel
```

min light	———●———	80
min lead	———●———	30

→

3 output

actual

red

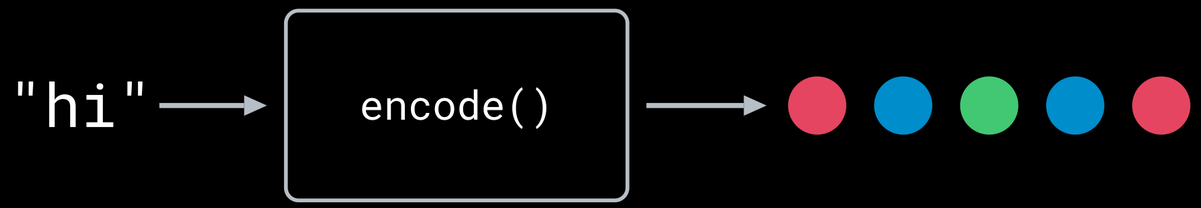
expected: `red` | `green` | `blue` | `off` | `uncertain`

✓ pass

`(203, 41, 57, 310) → red`

change the input or the rule. the test follows.

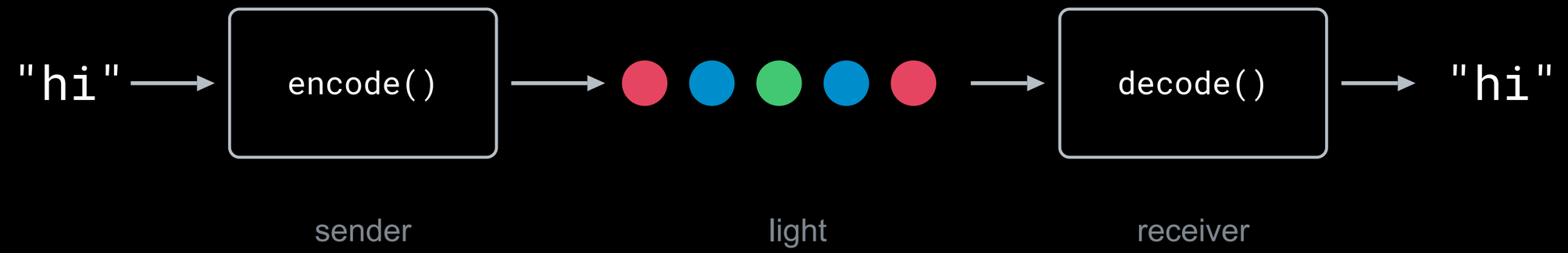
boxes connect when their edges agree



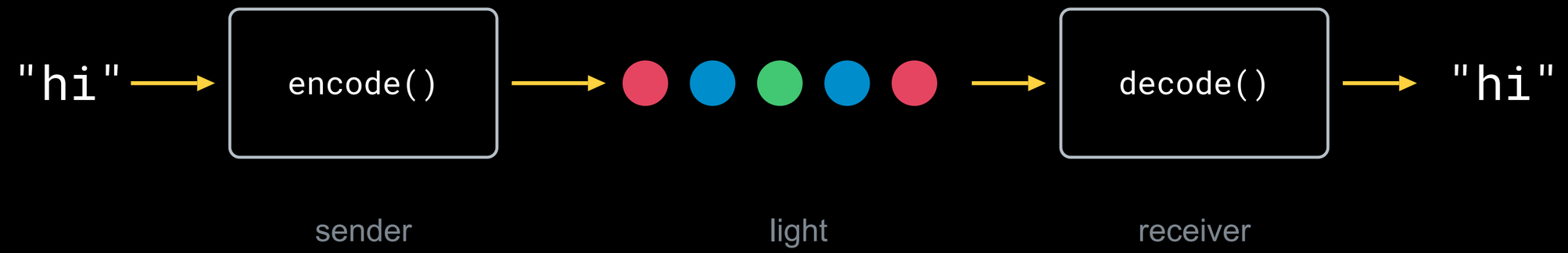
sender

light

boxes connect when their edges agree



boxes connect when their edges agree



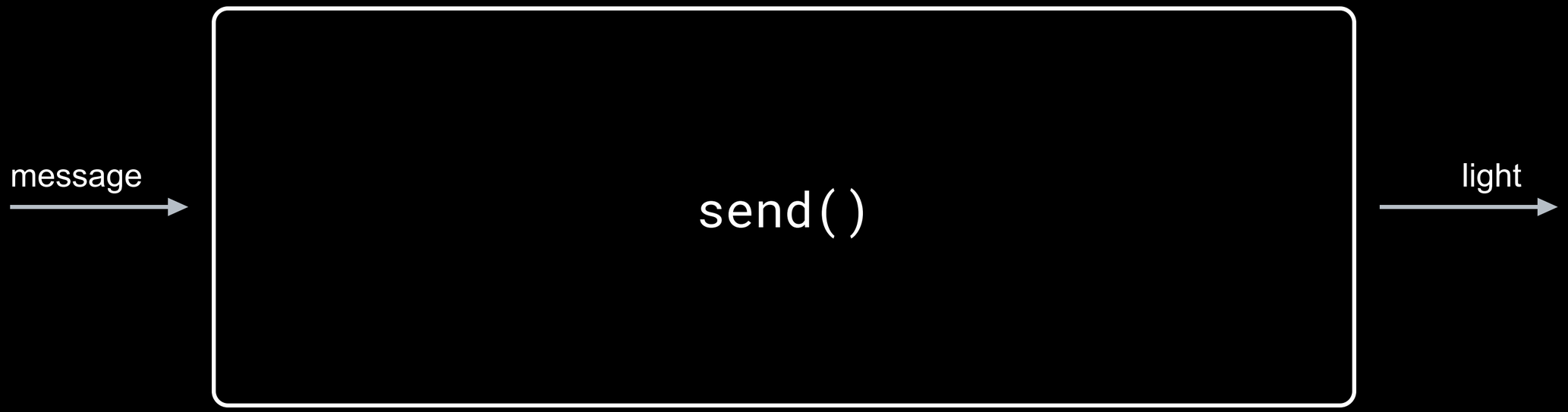
an agreement at every boundary.

to find the fault, follow the data.

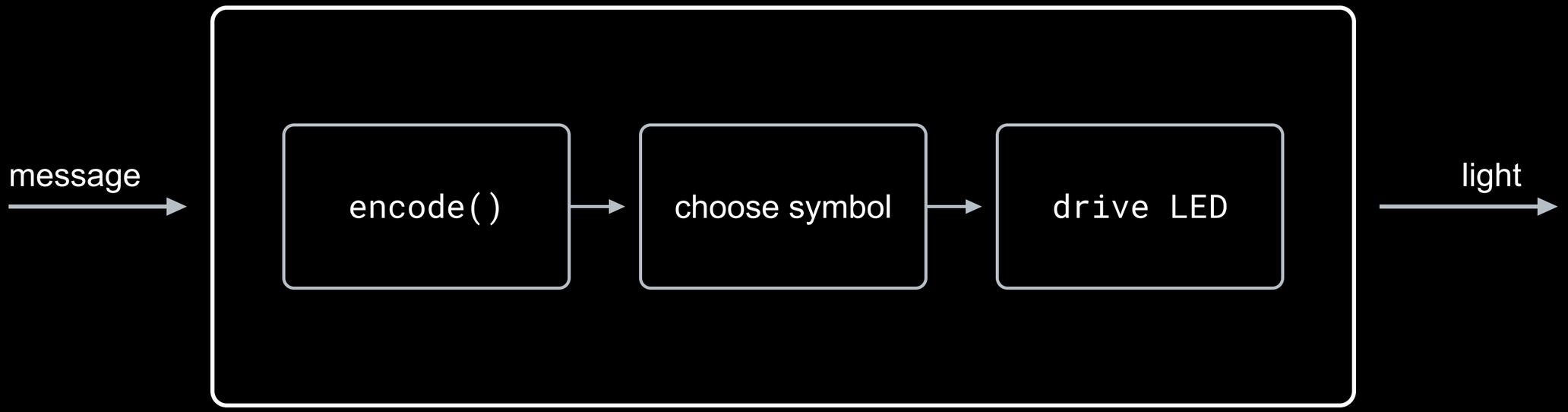
test one boundary at a time.

part 3

boxes all the way down

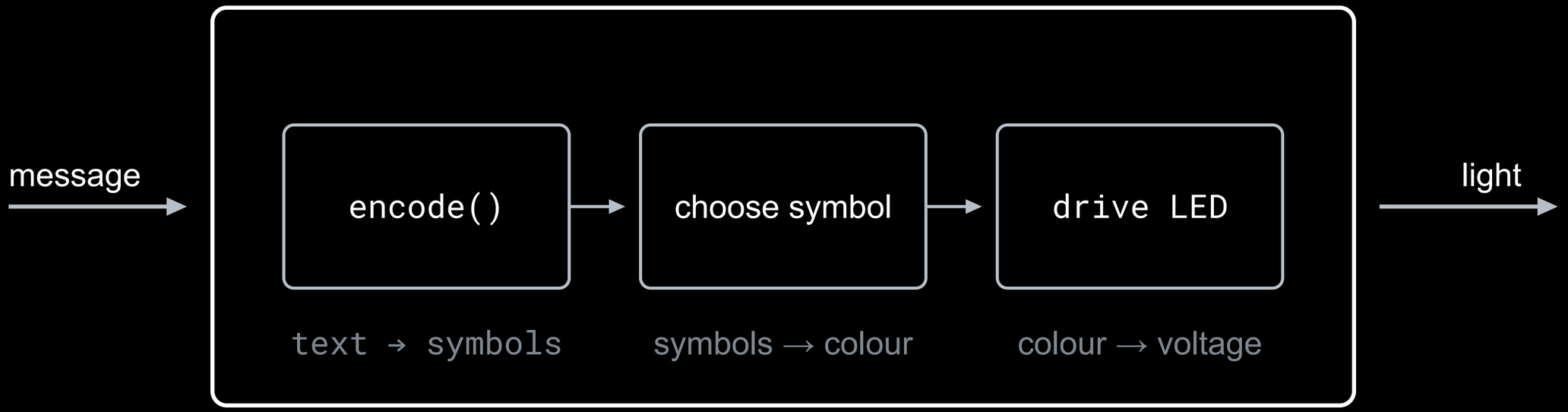


open a box only as far as your question requires.



open a box only as far as your question requires.

one box from the outside

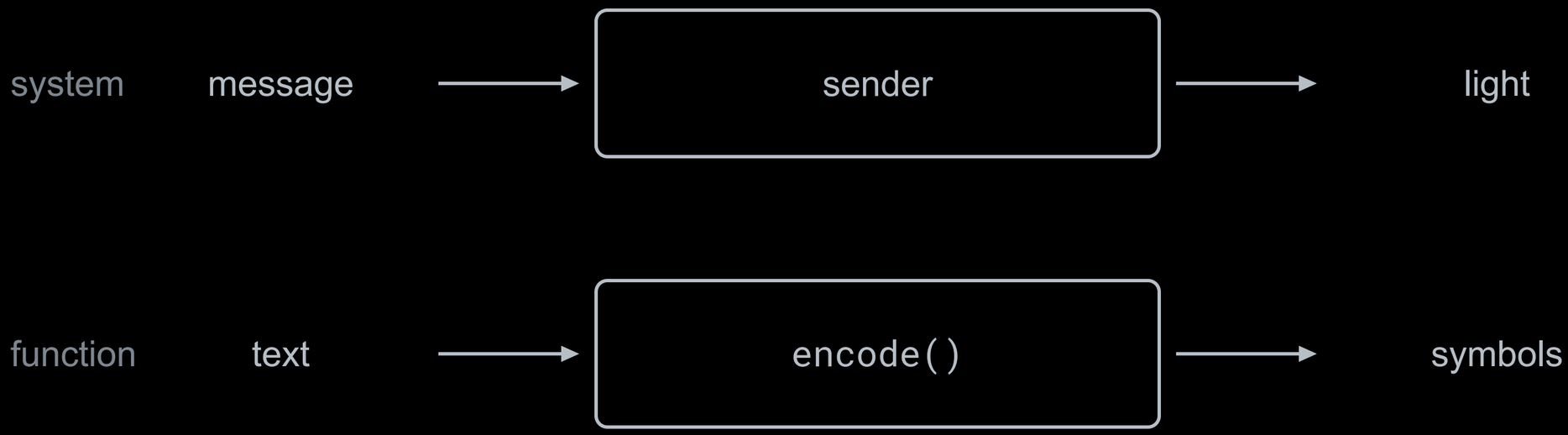


open a box only as far as your question requires.

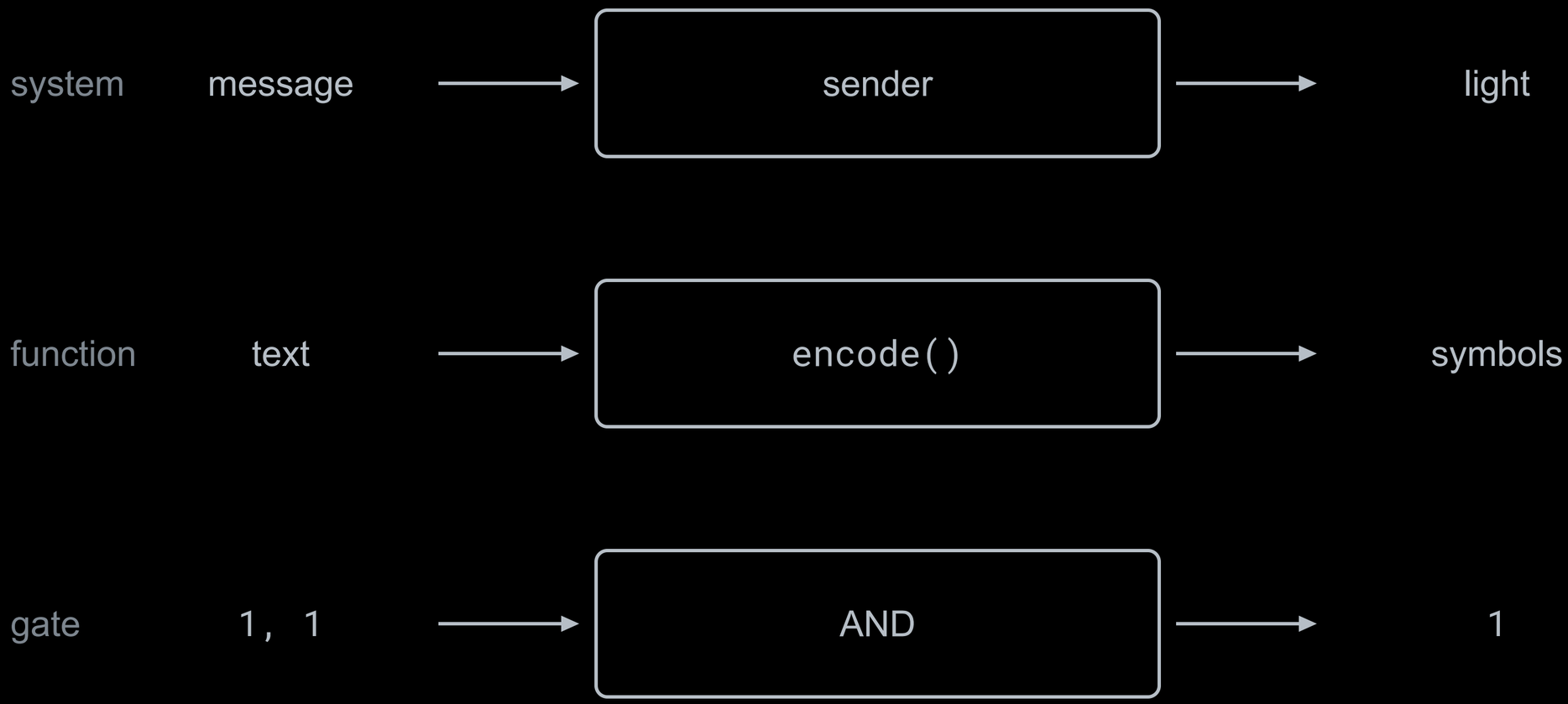
the shape survives every level



the shape survives every level

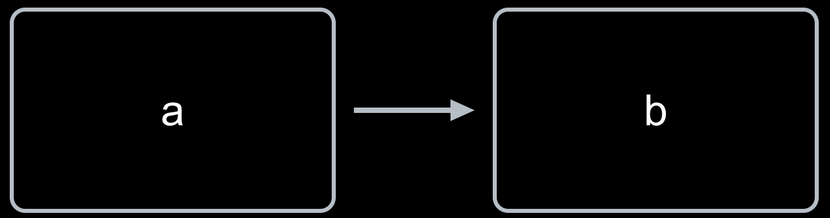


the shape survives every level



boxes in a row

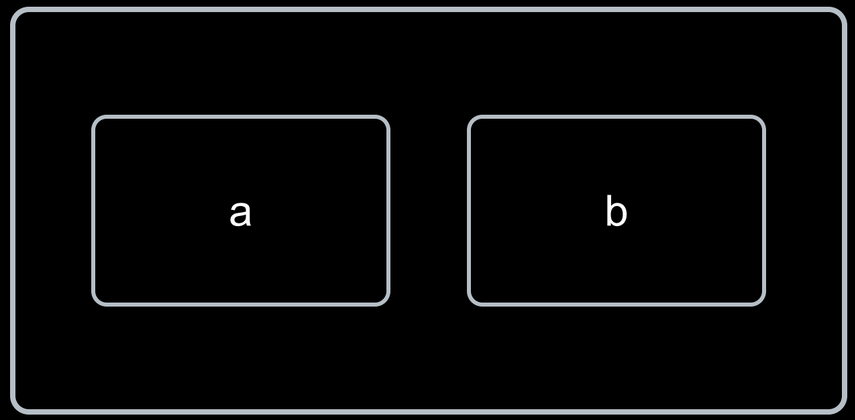
where do the data go next?



data flow

boxes inside boxes

what smaller work happens inside?



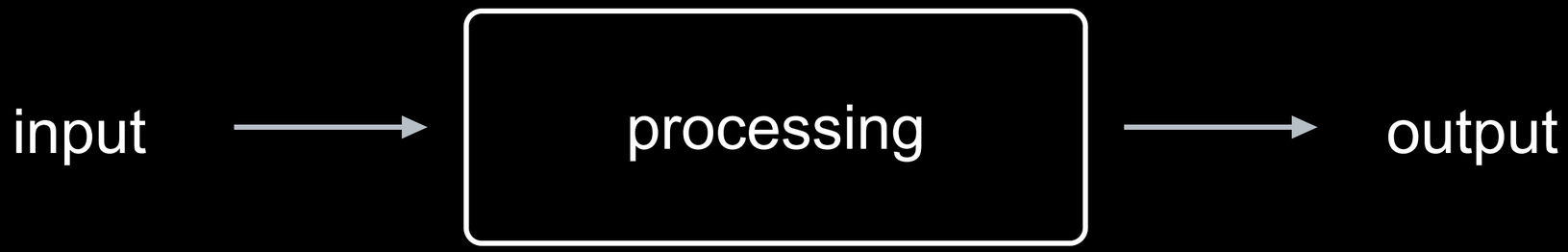
hierarchy

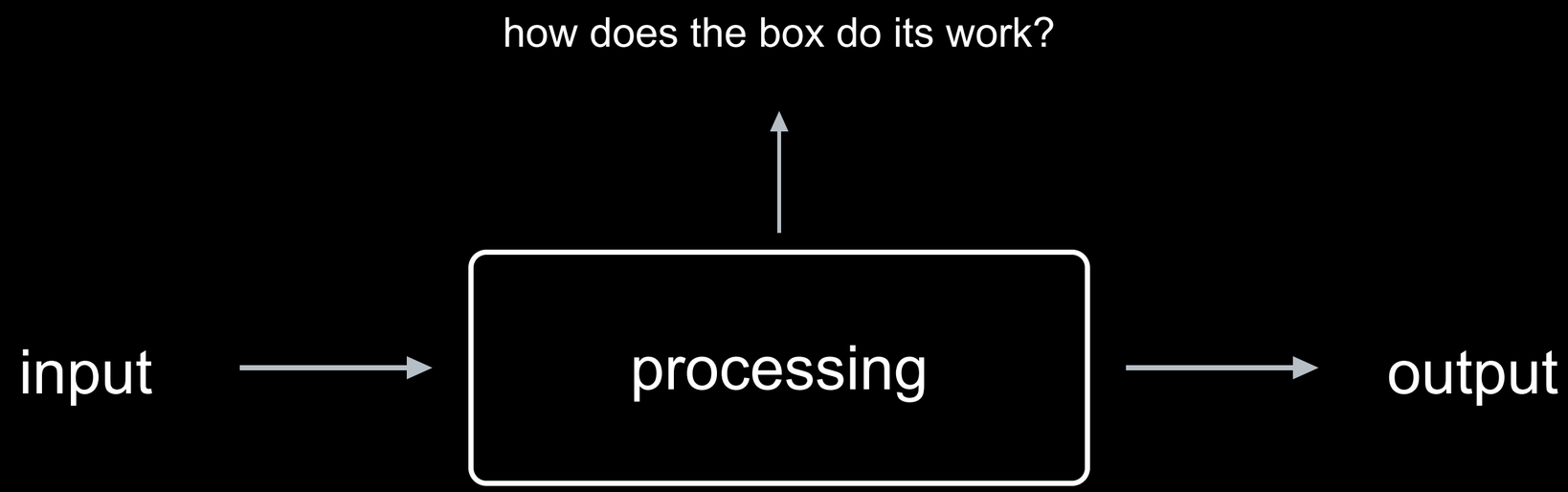
both are ways of cutting a problem, as in cutting problems: into steps in a row, or into smaller work inside.

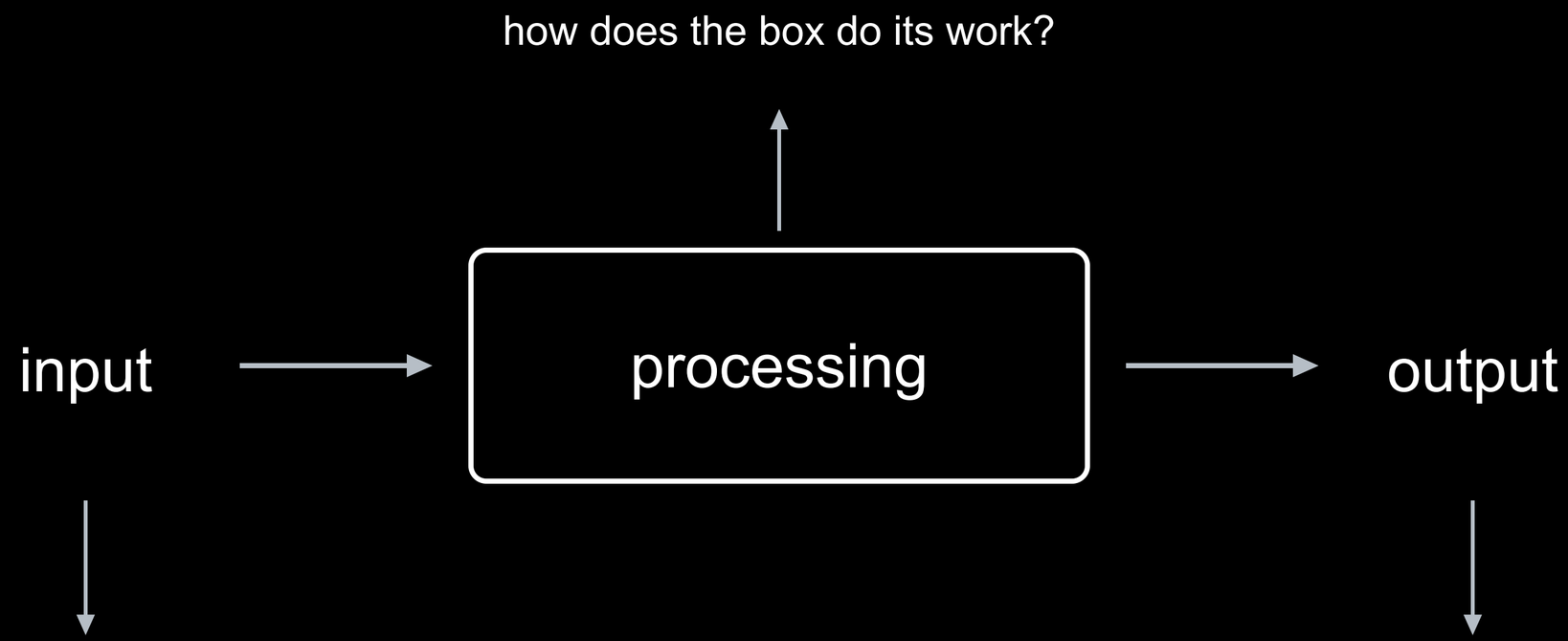
the right level is the one
you can test.

part 4

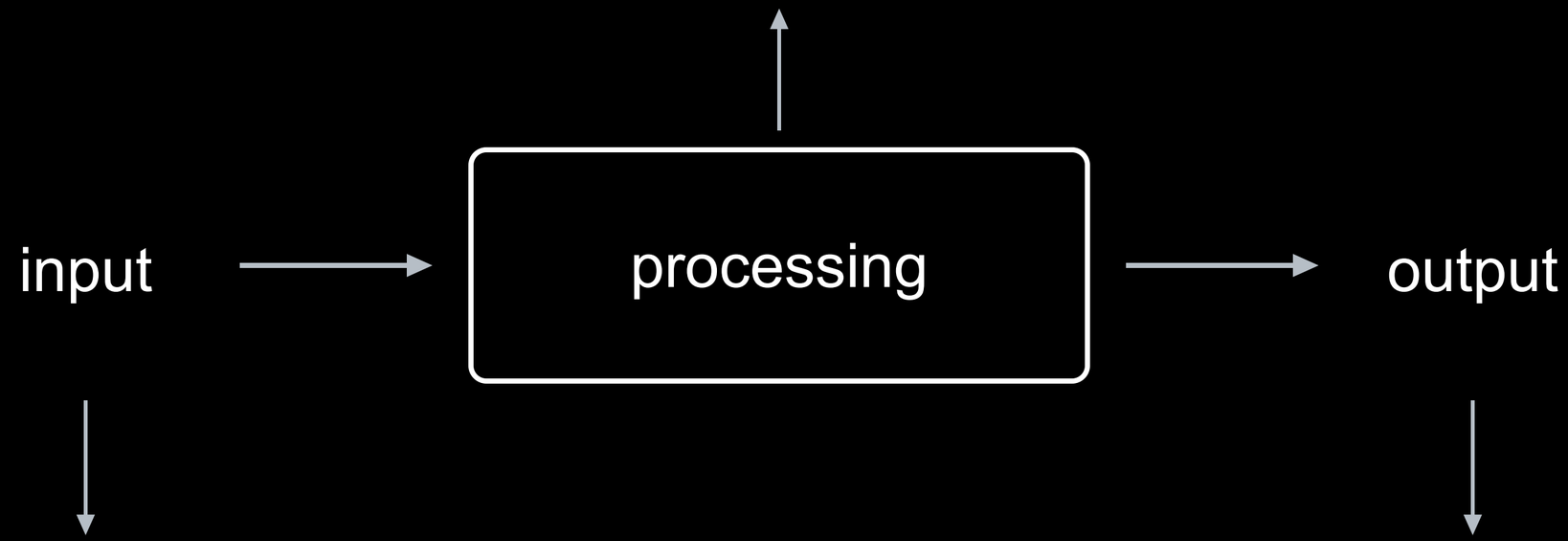
two questions inside every box


input
processing
output


how does the box do its work?
input
processing
output

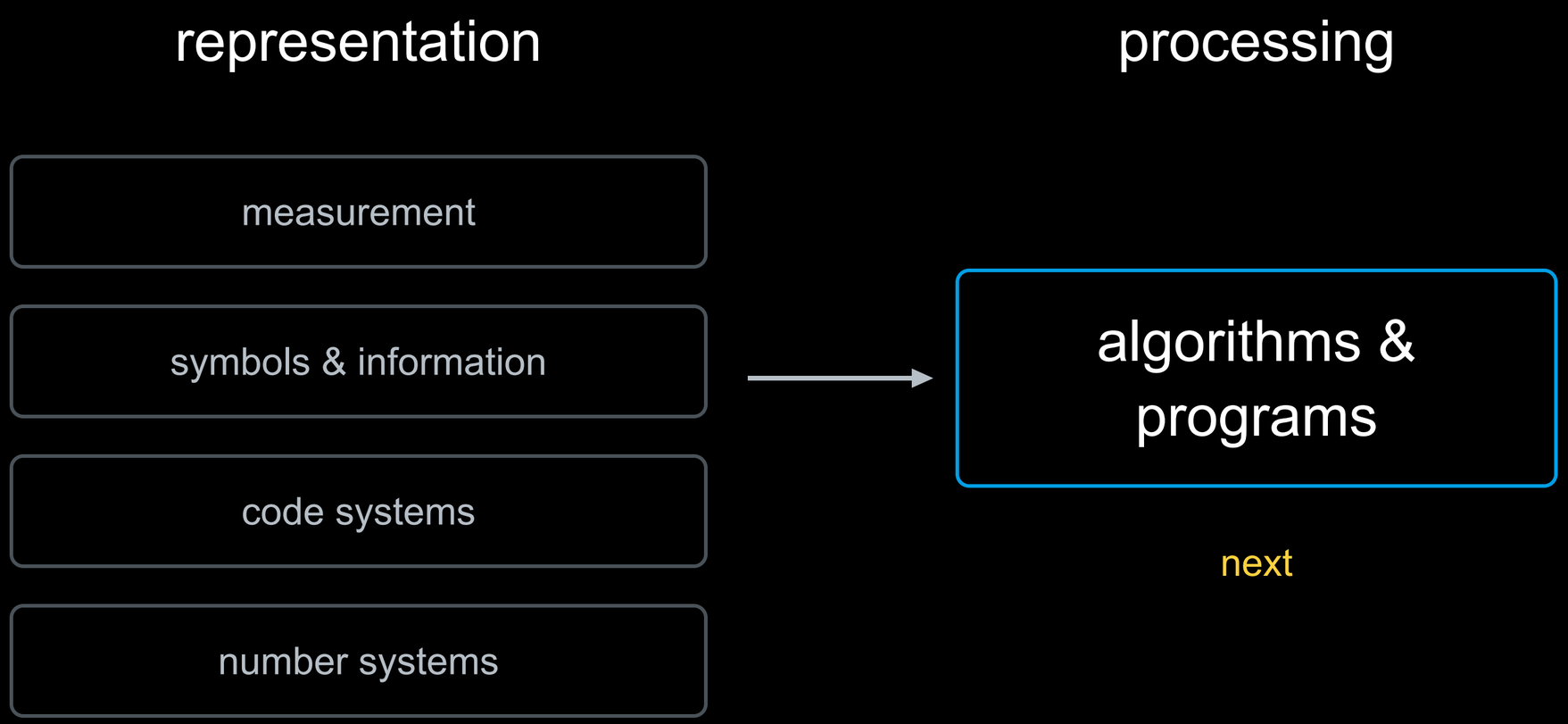

how does the box do its work?
input
processing
output
how are things written so a machine can work with them?

how does the box do its work? `program`



how are things written so a machine can work with them?

`representation`



the box stays. the question moves.

an AI agent is a box, too



you do not need trust when you have a test.



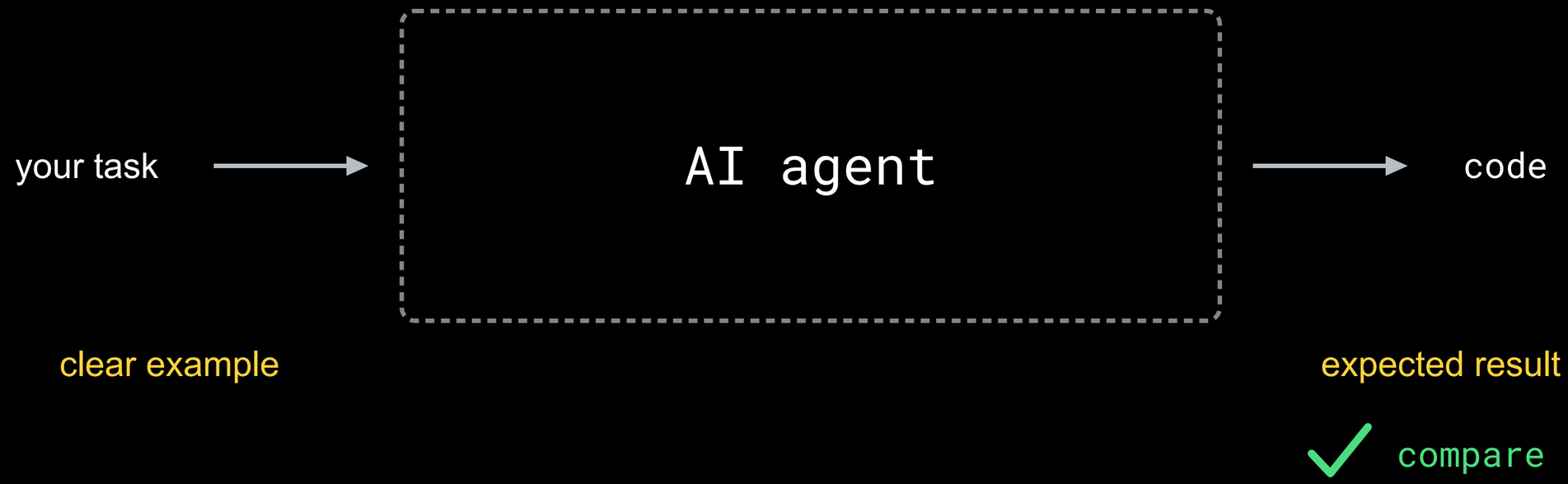
clear example

you do not need trust when you have a test.

an AI agent is a box, too



you do not need trust when you have a test.



you do not need trust when you have a test.

next:
open the box.



image: ai-generated (gpt-image-2)